Yilan Chen

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Research Summary

I develop algorithms for **LLM post-training**, focusing on reinforcement learning and on-policy distillation for reasoning models and coding agents. My research combines algorithm design and theoretical analysis with distributed LLM training and empirical evaluation. My recent work includes Reward-Aligned Distillation, which weights teacher supervision by its predicted effect on the student's task reward[1]; a framework connecting on-policy distillation to RL advantage estimation[2]; and self-distillation for agents using privileged information[3].

My broader research spans machine learning theory, including transformer in-context learning[8], provable dataset distillation[11], neural-network generalization[9][12][10][13], and connections between neural networks and kernel methods[14].

Education

University of California San Diego (UCSD) — San Diego, USA
Ph.D. in Computer Science — *Sep. 2022 – Mar. 2027 (expected)*

Advisors: Prof. Arya Mazumdar & Prof. Yian Ma
Committee: Mikhail Belkin, Sanjoy Dasgupta, Sicun Gao

University of California San Diego (UCSD) — San Diego, USA
M.S. in Computer Science — *Sep. 2020 – Jun. 2022*

Advisors: Prof. Tsui-Wei (Lily) Weng & Dr. Lam M. Nguyen (IBM Research)

Xi'an Jiaotong University (XJTU) — Xi'an, China
B.E. in Information Engineering — *Aug. 2015 – Jun. 2019*

Experience

Amazon AWS Agentic AI — Santa Clara, USA
LLM Post-training: Reinforcement Learning and Distillation — *Jun. 2026 – Sep. 2026*
Applied Scientist Intern · Hosts: Vijay Lingam, Rein Houthooft, Luke Huan

- Proposed **Reward-Aligned Distillation (RAD)**, a bilevel framework that weights token-level teacher supervision based on its predicted effect on the student's task reward, strengthening helpful signals and suppressing or reversing harmful ones. [paper]
- Built the end-to-end distributed LLM training pipeline in **verl** and developed a **custom Triton kernel** for attention Jacobian–vector products, enabling reward-guided token weighting during on-policy distillation.
- Improved mean accuracy across six **math benchmarks by 6.5 percentage points (pp)** for Qwen3-1.7B (non-thinking) and **LiveCodeBench-v6 pass@1 by 6.4 pp** for OLMo3-7B-Instruct over the strongest baselines.
- Applied RAD to **agentic software engineering**, achieving **50.5% pass@1 on SWE-bench Verified** with a Qwen3.5-4B student and Qwen3.6-27B teacher after half an epoch of R2E-Gym training, versus one epoch for the baselines.
- Contributed to the algorithm development of **PINT**, an on-policy **self-distillation** framework that internalizes privileged information into a temporary teacher's parameters, improving Qwen3-1.7B **WebShop success by 18.5 pp** over SDAR. [paper]

Google Research — New York, USA
On-Policy Distillation for LLM Reasoning — *Jun. 2025 – Sep. 2025*
Student Researcher · Hosts: Ankit Singh Rawat, Aditya Krishna Menon

- Developed a **theoretical framework connecting on-policy distillation with RL advantage estimation**, explaining how teacher models can supply dense token-level learning signals. (In submission)
- Proposed **discounted sequence- and token-level distillation methods** informed by generalized advantage estimation (GAE), and analyzed the bias–variance trade-offs of their gradient estimators.
- Implemented proposed algorithms in **TRL**, improving AIME 2025 accuracy by **9.4 pp over GRPO and 4.9 pp over KDRL**.
- Conducted pretraining, supervised fine-tuning, and knowledge-distillation experiments with Gemma 3 and Qwen3 models using Wikipedia, SlimPajama, Tulu3, and Skywork datasets.

The University of Tokyo and RIKEN AIP — Tokyo, Japan
Theory for Large Language Models and Deep Learning — *Jun. 2024 – Sep. 2024*
Research Intern · Advisor: Prof. Taiji Suzuki

- Developed theoretical analyses for emerging abilities of LLMs including **in-context learning, scaling laws, and chain-of-thought**.
- Established **sharp generalization bounds for gradient flow** through a data-dependent loss path kernel, combining stability analysis with Rademacher complexity (NeurIPS 2025).

University of California San Diego San Diego, USA
LLMs, Deep Learning Theory and Trustworthy Machine Learning *Jan. 2021 – Present*
Graduate Student Researcher · Advisors: Arya Mazumdar, Yian Ma, Lily Weng

- Analyzed **transformer in-context learning for single-index models**, characterizing optimization dynamics, optimal solutions, generalization, and scaling behavior.
- Developed a **provable and efficient dataset-distillation algorithm** for kernel ridge regression, establishing conditions under which **one synthetic example per class** exactly recovers the original model (NeurIPS 2024).
- Introduced the **loss path kernel** to connect neural-network loss dynamics with kernel methods, derive generalization bounds, and guide **neural architecture search** (NeurIPS 2023). Established **neural network–SVM equivalence** in the infinite-width neural tangent kernel regime (NeurIPS 2021).

Shanghai Jiao Tong University Shanghai, China
Interpretable Machine Learning *Jul. 2019 – Jun. 2020*
Research Intern · Advisor: Prof. Quanshi Zhang

- Developed quantitative measures of task-relevant and task-irrelevant visual concepts to explain the success of **knowledge distillation**, and evaluated the resulting hypotheses across neural-network architectures (CVPR 2020; TPAMI 2023).

Publications

* Equal contribution.

LLM Post-training, Agents & Distillation

[1] Reward-Aligned Distillation. [paper] [project]

Yilan Chen, Zian Su, Manish Prajapat, Dingmin Wang, Rein Houthooft, Rui Song, Vijay Lingam.
Under review at ICLR 2027; also accepted at NeurIPS 2026 PretoPost Workshop.

[2] Your RL Teacher Model Is Secretly a Value Network.

Yilan Chen, Ankit Singh Rawat, Aditya Krishna Menon, Veeranjaneyulu Sadhanala, Brian Hentschel, Shivam Gupta, Arya Mazumdar, Sanjiv Kumar.
Manuscript, 2026.

[3] PINT: Privileged Information Internalization for Agentic RL. [paper]

Zian Su, **Yilan Chen**, Manish Prajapat, Rein Houthooft, Rui Song, Vijay Lingam.
Under review at ACL 2027; also accepted at NeurIPS 2026 PretoPost Workshop.

[4] Token Utility for On-Policy Distillation via Local Trust-Region Optimization. [paper]

Narayanan Sadagopan, **Yilan Chen**, Vijay Lingam.
Under review at ICLR 2027; also accepted at NeurIPS 2026 PretoPost Workshop.

[5] Cross-Task Linearity Emerges in the Pretraining-Finetuning Paradigm. [paper] [code]

Zhanpeng Zhou*, Zijun Chen*, **Yilan Chen**, Bo Zhang, Junchi Yan.
ICML 2024.

[6] Quantifying the Knowledge in a DNN to Explain Knowledge Distillation for Classification. [paper]

Quanshi Zhang*, Xu Cheng*, **Yilan Chen**, Zhefan Rao.
IEEE TPAMI 2023.

[7] Explaining Knowledge Distillation by Quantifying the Knowledge. [paper]

Xu Cheng, Zhefan Rao*, **Yilan Chen***, Quanshi Zhang.
CVPR 2020.

Machine Learning Theory

[8] Multi-head Transformer In-context Learns Single-index Models.

Yilan Chen, Zhichao Wang, Hossein Taheri, Arya Mazumdar.
In submission.

[9] Generalization Bound of Gradient Flow through Training Trajectory and Data-dependent Kernel. [paper]

Yilan Chen, Zhichao Wang, Wei Huang, Andi Han, Taiji Suzuki, Arya Mazumdar.
NeurIPS 2025.

[10] How Does Label Noise Gradient Descent Improve Generalization in the Low SNR Regime? [paper]

Wei Huang*, Andi Han*, Yujin Song, **Yilan Chen**, Denny Wu, Difan Zou, Taiji Suzuki.
NeurIPS 2025.

[11] Provable and Efficient Dataset Distillation for Kernel Ridge Regression. [paper] [code] [project]

Yilan Chen, Wei Huang, Tsui-Wei Weng.
NeurIPS 2024.

[12] Analyzing Generalization of Neural Networks through Loss Path Kernels. [paper] [code]

Yilan Chen, Wei Huang, Hao Wang, Charlotte Loh, Akash Srivastava, Lam M. Nguyen, Tsui-Wei Weng.
NeurIPS 2023.

[13] Analyzing Deep PAC-Bayesian Learning with Neural Tangent Kernel: Convergence, Analytic Generalization Bound, and Efficient Hyperparameter Selection. [paper]

Wei Huang*, Chunrui Liu*, **Yilan Chen**, Richard Yi Da Xu, Miao Zhang, Tsui-Wei Weng.
TMLR 2023.

[14] On the Equivalence between Neural Network and Support Vector Machine. [paper] [code]

Yilan Chen, Wei Huang, Lam M. Nguyen, Tsui-Wei Weng.
NeurIPS 2021.

Other Publications

[15] BridgeSim: Unveiling the OL-CL Gap in End-to-End Autonomous Driving. [paper]

Seth Z. Zhao*, Luobin Wang*, Hongwei Ruan, Yuxin Bao, **Yilan Chen**, et al.
CoRL 2026.

[16] The Importance of Prompt Tuning for Automated Neuron Explanations. [paper] [code]

Justin Lee*, Tuomas Oikarinen*, Arjun Chatha, Keng-Chi Chang, **Yilan Chen**, Tsui-Wei Weng.
NeurIPS 2023 ATTRIB Workshop.

Invited Talks

Generalization Bound of Gradient Flow through Training Trajectory and Data-dependent Kernel
EnCORE Institute Workshop on Theoretical Perspectives on LLMs — *Mar. 2025*

Analyzing Neural Networks through Equivalent Kernels
RIKEN AIP–SJTU CS Joint Workshop on Machine Learning and Brain-like Intelligence — *Aug. 2024*
Southern California Applied Mathematics Symposium (SOCAMS 2024) — *Apr. 2024*

Analyzing Generalization of Neural Networks through Loss Path Kernels
ByteDance — *Jan. 2024*
AI TIME — *Nov. 2023*

Teaching

DSC 212: Probability and Statistics for Data Science, Teaching Assistant — *Fall 2024*
DSC 140B: Representation Learning, Teaching Assistant — *Spring 2024*
DSC 210: Numerical Linear Algebra, Teaching Assistant — *Fall 2023*
DSC 291: Trustworthy Machine Learning, Teaching Assistant — *Fall 2021*

Professional Service

Conference Reviewer: ICML (2022–2025), NeurIPS (2022), ICLR (2022, 2024–2026)
Area Chair: ICLR 2026 DeLTa Workshop
Journal Reviewer: Journal of Optimization Theory and Applications (JOTA)

Technical Skills

Programming: Python, C/C++, MATLAB, JavaScript
Machine Learning & Systems: PyTorch, JAX, Verl, TRL, distributed training, Triton, Linux
Theory: Linear algebra and matrix analysis, high-dimensional statistics, (non-)convex optimization

Honors & Awards

NeurIPS 2023 Scholar Award — *Dec. 2023*
Simons Institute Deep Learning Theory Workshop Funding — *Aug. 2022*
Outstanding Student Award — *Sep. 2016 and 2018*
"Siyuan" Merit Scholarship — *Sep. 2016, 2017, and 2018*